

Lab 1: Your First Reproducible R Script

Level: Foundation Mapping: LO1 | K4 | A2 Concepts: objects, assignment, Source vs Console, comments, help

Scenario

A training coordinator needs a repeatable summary of course attendance.

Goal

A saved .R script that creates, summarises and exports attendance data.

Files in this folder

- lab-01.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Open the workbook and review the Brief and Input_Data sheets.
2. Create lab-01.R in RStudio and add a header comment with purpose and source.
3. Create the attendance vector and calculate valid count, mean and minimum.
4. Use `help('mean')` and record the `na.rm` decision.
5. Export the summary and run `verify.R`.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-01.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-01.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open `analysis_output.csv` and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Script runs from a clean R session without manual Console steps.
- [] Summary excludes the missing value and reports the expected valid count.
- [] Evidence sheet records the output filename and verification result.

Troubleshooting

- there is no package called 'readxl': the script automatically uses `input-data.csv`; install `readxl` later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.